FIITJEE

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Main)-2020

TEST DATE: 29-12-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. B

Sol. Maximum range up the inclined plane = $\frac{u^2}{g(1+\sin\theta)}$

Maximum range down the inclined plane = $\frac{u^2}{g(1-\sin\theta)}$

The maximum possible distance between the two bullets after they hit the inclined plane

$\therefore\ d_{max} = \frac{u^2}{g(1+\sin\theta)} + \frac{u^2}{g(1-\sin\theta)}$

$d_{max} = \frac{2u^2}{g\cos^2\theta}$

2. B

Sol. Use the basic concept of interference of light waves.

3. C

Sol. Acceleration of block = $-3\ m/s^2$

Acceleration of plank = $-7\ m/s^2$

After 3 sec, the plank will come to rest. In 3 sec, the block will move $\frac{1}{2}\times 4\times 3\times 3 = 18$ m relative to the plank.

Now, let the block will cover remaining 18 m relative to plank in next t seconds.

$12t - \frac{1}{2}\times 3t^2 = 18$

$8t - t^2 = 12 \Rightarrow t = 2$ sec

Total time after which the block will get off the plank, $t_0 = 3 + 2 = 5$ sec.

4. B

Sol. At θ_{min}

$\lambda gR(1-\sin\theta) - \lambda gR(1-\cos\theta) - \mu\lambda gR(\sin\theta + \cos\theta) = 0$

$\Rightarrow \cos\theta - \sin\theta - \frac{\sin\theta}{3} - \frac{\cos\theta}{3} = 0$

$\therefore \theta = \tan^{-1}(1/2)$

5. B

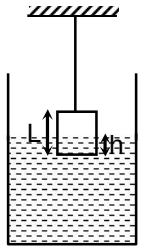
Sol. $f \propto \sqrt{T}$

$T_0 = n\rho_w L^3 g$

$T_{new} = n\rho_w L^3 g - 2\rho_w L^2 hg = \rho_w L^2 g(nL - 2h)$

$\therefore \frac{f_{new}}{f_0} = \sqrt{\frac{\rho_w L^2 g(nL-2h)}{n\rho_w L^3 g}}$

$f_{new} = \left(\frac{nL-2h}{nL}\right)^{1/2} f_0$



6. C

Sol. $\phi = \frac{\pi}{2}$

$\frac{t}{T} = \frac{\pi/2}{2\pi}$

$\therefore t = T/4 = \frac{\pi}{2}\sqrt{\frac{m}{k}}$

7. D

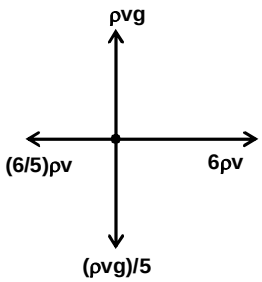
Sol. $3L(1+\alpha_{eff}\Delta\theta) = L(1+2\alpha\Delta\theta) + 2L(1+\alpha\Delta\theta)$

$3L\alpha_{eff}\Delta\theta = 4\alpha L\Delta\theta$

$\therefore \alpha_{eff} = \frac{4}{3}\alpha$

8. A

Sol. F.B.D.



9. A

Sol. $I = neAv_d$

$I = kV^2$

10. D

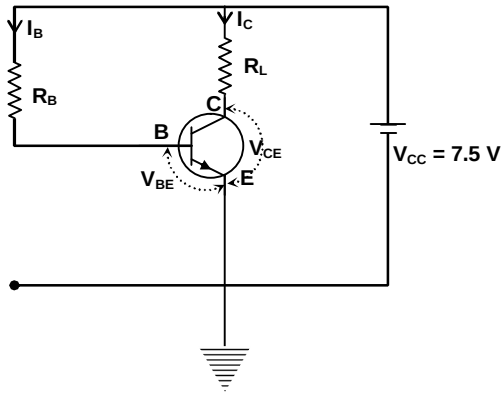
Sol. $V_{CC} = I_BR_B + V_{BE}$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{7.5 - 1}{130 \times 10^3} = 50 \times 10^{-6} = 50\mu A$$

$$I_C = \beta I_B = 100 \times 5 \times 10^{-5} = 5 \times 10^{-3} A = 5\ mA$$

Now, $I_CR_L + V_{CE} = V_{CC}$

$$R_L = \frac{V_{CC} - V_{CE}}{I_C} = \frac{7.5 - 3.5}{5 \times 10^{-3}} = \frac{4000}{5} = 800\Omega$$



11. A

Sol. d(K.E.) = mv× Δv

Change in velocity Δv has fixed magnitude

Greatest change in kinetic energy is achieved when impulse is parallel to the velocity and when the speed is as large as possible.

12. C

Sol. Then intensity of the emerging light, $I = \frac{I_0}{2}\cos^2(37°) = \frac{I_0}{2} \times \left(\frac{4}{5}\right)^2$

$$I = \frac{8I_0}{25} = 8\ W/m^2$$

13. C

Sol. $V_0 = \frac{k\rho a^2}{\varepsilon_0}$ = potential at the centre of a cube of side 'a'

$k\rho \times \frac{4a^2}{\varepsilon_0} = 4V_0$ = potential at the centre of a cube of side '2a'

Six such pyramids make a cube

Potential at the tip of pyramid = $\frac{4V_0}{6} = \frac{2V_0}{3}$

14. B

Sol. $\frac{dQ_1}{T_1} + \frac{dQ_2}{T_2} = 0$

$\frac{dT_1}{T_1} = -\frac{dT_2}{T_2} \Rightarrow T_1T_2$ = constant

When their temperatures become equal

$T_1T_2 = T^2$

$\Rightarrow$ Final temperature, $T = \sqrt{T_1T_2}$

15. A

Sol. t = 1.5 h

Total charge transferred = 4.5 Ah

Q_{max} = 60 Ah

Percentage charged = $\frac{4.5}{60} \times 100 = 7.5\%$

16. C
Sol. Use Faraday's Law

17. A
Sol. $nh\nu = \frac{100 \times 60}{100} = 60$ J

$n = \frac{60\lambda}{hc} = \frac{60 \times 3.9 \times 10^{-7}}{6.63 \times 10^{-34} \times 3 \times 10^{8}} = 1.18 \times 10^{20}$

18. A
Sol. Using conservation of momentum
$mu = mv_1 + Mv_2$
$\Rightarrow 2v_2 + v_1 = u$ …(i)
$v_2 - v_1 = u$ …(ii)
from (i) and (ii)
$v_2 = 2u/3$

$\therefore$ Fraction of kinetic energy lost by neutron= $\frac{\frac{1}{2} \times 2m \times \left(\frac{2u}{3}\right)^2}{\frac{1}{2}mu^2} = \frac{8}{9}$

19. C
Sol. $2 \times \frac{1}{2g}\left(\frac{p\sin\theta}{m}\right)^2 = \frac{2p\sin\theta}{mg} \times \frac{p\cos\theta}{m}$

$\frac{1}{2}\sin^2\theta = \sin\theta\cos\theta \Rightarrow \tan\theta = 2$

$\therefore \cos\theta = \frac{1}{\sqrt{5}}$

Minimum kinetic energy = $\frac{(p\cos\theta)^2}{2m} = \frac{p^2}{2m} \times \frac{1}{5} = \frac{p^2}{10m}$

20. D
Sol. $A = A_0 e^{-\frac{bt}{2m}}$

$A = A_0 e^{-\frac{\ell n2}{10 \times 2} \times \frac{20}{2}} = A_0 e^{-\ell n\sqrt{2}} = \frac{A_0}{\sqrt{2}}$

SECTION – B

21. 5
Sol. $H = B/\mu_0$
$ni = 2 \times 10^4$
$40 \times 100\, i = 2 \times 10^4$
$i = \frac{20}{4} A = 5A$

22. 7
Sol. $m_1 g - T_1 = m_1 a$ …(i)
$T_2 - m_2 g = m_2 a$ …(ii)

$T_1 - T_2 \pm \frac{m_1 g}{2} = m_1 a$ …(iii)

$(m_1 - m_2)g \pm \frac{m_1 g}{2} = (2m_1 + m_2)a$

$a = \frac{2m_2 g \pm \frac{3gm_2}{2}}{7m_2}$

$a_{max} = \frac{g}{2}$

$a_{min} = \frac{g}{14}$

$\frac{a_{max}}{a_{min}} = 7$

SECTION – C

23. 00004.80

Sol. $(\mu_S - 1)t_0 = \frac{d \times y}{D}$

$y = (\mu_S - 1)t_0 \frac{D}{d}$

$\Rightarrow$ velocity of central maxima,

$v = \frac{dy}{dt} = (2t + 12) \times \frac{2.4 \times 10^{-4} \times 2}{2 \times 10^{-3}}$

At t = 4 sec

$v = 20 \times 2.4 \times 10^{-1} = 4.80$ m/s

24. 00086.60

Sol. $N = mg\cos\alpha - F\sin\theta$

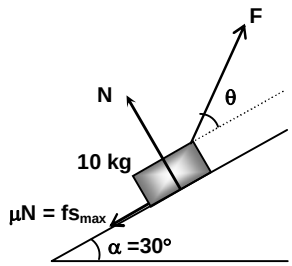
$F\cos\theta = \mu N + mg\sin\alpha$

$F(\cos\theta + \mu\sin\theta) = mg(\sin\alpha + \mu\cos\alpha)$

$F = \frac{mg(\sin\alpha + \mu\cos\alpha)}{(\cos\theta + \mu\sin\theta)}$

$\therefore F_{min} = \frac{mg(\sin\alpha + \mu\cos\alpha)}{\sqrt{1+\mu^2}}$ [when $\theta = \tan^{-1}(\mu) = 30°$]

$\therefore F_{min} = 86.60$ N



25. 00000.25

Sol. $\frac{5\lambda}{4} = \ell_1 + e$ …(i)

$\frac{9\lambda}{4} = \ell_2 + e$ …(ii)

From (i) and (ii)

$\lambda = \ell_2 - \ell_1 = 200 - 111 = 89$ cm

From (i), end correction

$e = \frac{5\lambda}{4} - \ell_1 = \frac{5}{4} \times 89 - 111$

$\Rightarrow e = 0.25$ cm

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

26. A

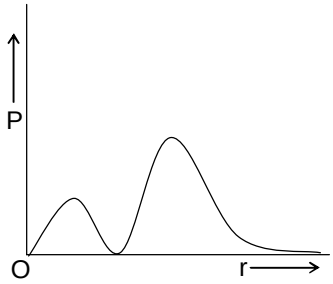
Sol. $H_3C-\underset{\underset{Cl}{|}}{CH}-CH_3$ will respond the test.

27. A

Sol. Aqueous solution of $CH_3-\overset{\overset{O}{\|}}{C}-OH$ and $CH_3-\overset{\overset{O}{\|}}{C}-ONa$ is a buffer solution.

28. D

Sol. Sketch for 2s orbital is



29. A

Sol. E_{cell} is an intensive property, where as mass, internal energy, heat capacity are extensive properties.

30. C

Sol. $dE = dq + dW$

$$dE + PdV = dq \Rightarrow \frac{dE}{dT} + P\frac{dV}{dT} = \frac{dq}{dT}$$

$$\Rightarrow nC_{V,m} + \frac{PdV}{dT} = \frac{dq}{dT}$$

$$\Rightarrow 1\times\frac{fR}{2} + \frac{PdV}{dT} = \frac{dq}{dT}$$

$$\Rightarrow \frac{5R}{2} + \frac{PdV}{dT} = \frac{dq}{dT} \quad \ldots. (1)$$

$$PV = RT \Rightarrow V^3 = RT$$

$$\Rightarrow 3V^2\frac{dV}{dT} = R \Rightarrow \frac{PdV}{dT} = \frac{R}{3}$$

$$\frac{5R}{2} + \frac{R}{3} = \frac{dq}{dT}$$

$$\frac{17R}{6} = \frac{dq}{dT}$$

31. A
Sol. Forsterite [$Mg_2(SiO_4)$] is orthosilicate.

32. B
Sol. Given complex have four possible coordination isomers.

en	CN	CN	en
3	$0]^{3+}$	6	$0]^{3-}$
2	$2]^{1+}$	4	$1]^{1-}$
1	$4]^{-1}$	2	$2]^{+1}$
0	$6]^{-3}$	0	$3]^{3+}$

$\left[Co(en)_3\right]\left[Cr(CN)_6\right]$

$\left[Co(en)_2(CN)_2\right]\left[Cr(CN)_4(en)\right]$

$\left[Co(en)_1(CN)_4\right]\left[Cr(CN)_2(en)_2\right]$

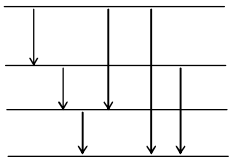
$\left[Co(CN)_6\right]\left[Cr(en)_3\right]$

33. A
Sol. Number of streo isomers possible for the given complex is 8.

34. A
Sol. P_4O_{10} is not an oxidizing agent.

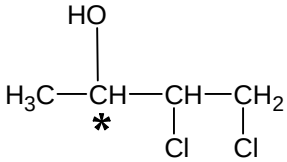
$P_4O_{10} + NaCl \longrightarrow POCl_3 + Na_3PO_4$

35. A
Sol. Minimum 4 hydrogen atom should be present in sample to get 6 different spectral lines.



36. B
Sol. Barytes is a sulphate ore.

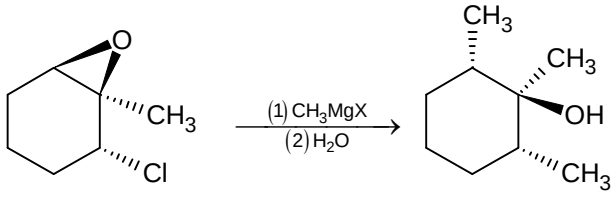
37. C
Sol. Two optical isomers are possible for the given organic molecule



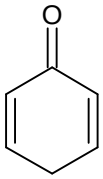
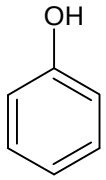
38. A
Sol. Total 5 dihalogen derivatives are formed including stereoisomers in which two are diastereomers.
$\therefore x + y = 10$.

39. C

Sol.

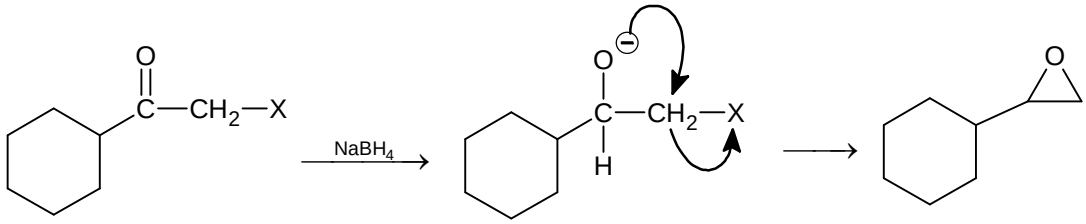


40. C

Sol.  will tautomerise to form  .

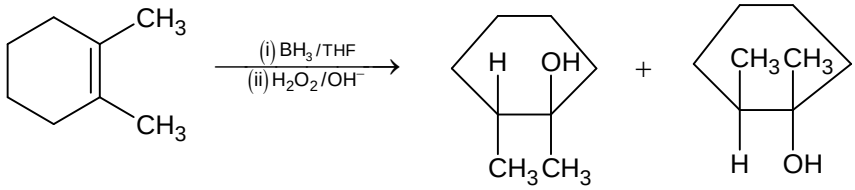
41. B

Sol.



42. D

Sol.



Number of products = 2.

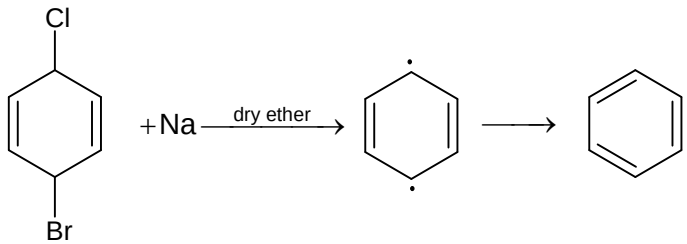
43. C

Sol. $$Z = \frac{PV_m}{RT} = \frac{V_m}{V_m - b} = \frac{11b}{11b - b} = \frac{11}{10}$$

$$\because \frac{P(V - nb)}{n} = RT \Rightarrow \frac{RT}{P} = \frac{V - nb}{n} = V_m - b$$

44. B

Sol.



45. B

Sol. $B + 3D \longrightarrow 4A + 4C$

Rate of reaction $= \frac{-d[B]}{dt} = \frac{-1}{3}\frac{d[D]}{dt} = \frac{1}{4}\frac{d[A]}{dt} = \frac{1}{4}\frac{d[C]}{dt}$

SECTION – B

46. 1

Sol. $Be_2^+ : \sigma 1^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^1$

B. O. = $\frac{1}{2}[4-3] = 0.5$

$\Rightarrow 2x = 1$

47. 4

Sol. Following are dibasic acid

H_2SO_4, H_3PO_3, H_2SO_5 and $H_2S_2O_7$.

SECTION – C

48. 00200.00

Sol. Let the final composition, liquid (10 mole)

A = x mole, B = (10 – x) mole

Vapour (10 moles), A = 10 – x mole; B = x mole

Now, $\frac{Y_A}{Y_B} = \frac{X_A}{X_B}\frac{P_A^o}{P_B^o} \Rightarrow \frac{10-x}{x} = \frac{x}{10-x} \times \frac{400}{100}$

$x = \frac{10}{3}$

Now, $P = X_A P_A^o + X_B P_B^o$

$= \frac{x}{10} \times 400 + \frac{10-x}{10} \times 100$

= 200.00

49. 00085.40

Sol. $K_2Cr_2O_7 + 14HCl \longrightarrow 2KCl + 2CrCl_3 + 3Cl_2 + 7H_2O$

n-factor of $K_2Cr_2O_7$ = 6 and

n-factor of HCl $= \frac{3}{7}$

$\therefore E_{HCl} = \frac{M}{\frac{3}{7}} = \frac{7M}{3} = 85.40$

= 85.40

50. 00006.50

Sol. pH at isoelectric point = $\frac{pKa_1 + pKa_2}{2} = \frac{4+9}{2} = 6.5$

Mathematics

PART – III

SECTION – A

51. B

Sol. Area is minimum when $c = \frac{-3+9}{2} = 3$

52. B

Sol. $\frac{25!\,(-1)^2}{22!\,2!\,1!} = 6900$

53. A

Sol. $b + 3 = \pm 4$, $b = -7, 1$

54. A

Sol. Coefficient of x° in $(x^{-3} + x^{-2} + \ldots.. + x^4)^3 - 1 = {}^{11}C_9 - 3 \times {}^3C_1 - 1 = 46 - 1 = 45$

55. B

Sol. Number of even divisor of 80 is 8

56. B

Sol. $k = 2$, $a > 3$

57. A

Sol. $x + y = 47$, number of matrix = ${}^{46}C_1 = 46$

58. D

Sol. $\alpha = -\sqrt{\frac{17}{2}}$, $\beta = 0$, $\gamma = \sqrt{\frac{17}{2}}$

59. A

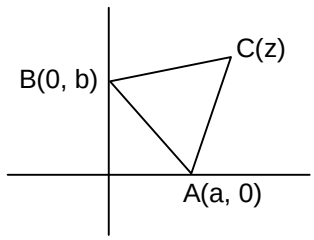
Sol. Using graph clearly see
f(x) is continuous everywhere but not differentiable at exactly 3 points

60. A

Sol. m = 10 and n = 12

61. A

Sol. Let A(a, 0), B(0, b)



$\frac{z-a}{z-ib} = e^{i\theta} = \frac{3}{5} + i\frac{4}{5}$

$5(z - a) = (z - ib)(3 + 4i)$

$5(x - a + iy) = (x + iy - bi)(3 + 4i)$

Comparing, we get $5(x - a) = 3x - 4(y - b)$

$2x + 4y = 5a + 4b$ ..... (1)

$5y = 4x + 3(y - b)$

$4x - 2y = 3b$ ..... (2)

$2x + 4y = 5a + \frac{4(4x - 2y)}{3} \Rightarrow 2x - 4y + 3a = 0$

62. D

Sol. $\frac{1}{\ell_1}+\frac{1}{\ell_2}=\frac{1}{4a}$

63. A

Sol. $a^2=\frac{5}{2},\ b^2=\frac{5}{12}$

64. D

Sol. $P(x) = (x + 3)(x - 3)^3 + x^2 + 2$

65. B

Sol. Radius of circle = $\sqrt{34}$

66. C

Sol. Required probability = $\frac{91^{10}-82^{10}}{91^{10}}$

67. C

Sol. It is neither tautology nor contradiction

68. D

Sol. Data are multiplied by 7, then variance should be 49 times

69. A

Sol. $tx = y \Rightarrow dt=\frac{dy}{x}\ \therefore\ \frac{1}{x}\int_0^x f(y)dy = nf(x)\ \therefore\ \int_0^x f(y)dy = x\cdot n\cdot f(x)$

Differentiating $f(x) = n(f(x) + xf'(x)) \Rightarrow \frac{f'(x)}{f(x)}=\frac{1-n}{nx}$

Integrating, $\ln f(x)=\left(\frac{1-n}{n}\right)\ln(cx)\ \Rightarrow f(x)=(cx)^{\left(\frac{1-n}{n}\right)}\ \Rightarrow f(x)=c\cdot(x)^{\left(\frac{1-n}{n}\right)}$

70. D

Sol. $\lambda = 0 \Rightarrow x + y = \frac{\pi}{4} \Rightarrow$ Equation of mirror image is $x - y = \frac{\pi}{4}$

SECTION – B

71. 1

Sol. $A = -1,\ B = 1,\ f\left(-\frac{\pi}{4}\right)=-1$

72. 1

Sol. Use A.M. $\geq$ G.M.

SECTION – C

73. 00001.75

Sol. $\theta=\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)=\sin^{-1}\left(\frac{2}{\sqrt{7}}\right)\ \Rightarrow \frac{a}{2b}=\frac{7}{4}=1.75$

74. 00000.44

Sol. $\frac{x-3}{1}=\frac{y-2}{2}=\frac{z-0}{1}=-2\left(\frac{3+4+1}{1+4+1}\right)=\frac{-8}{3}$; $x=\frac{1}{3}$; $y=\frac{-10}{3}$; $z=\frac{-8}{3}$; $\frac{\lambda}{3}-\frac{10}{3}+3\lambda\left(\frac{8}{3}\right)+7=0$

$25\lambda = -11$, $\lambda=\frac{-11}{25}$

75. 00001.65

Sol. $\frac{14\times\frac{12!}{5!\times 2!}}{\frac{7!}{2!}\times {}^{8}P_{5}}=1.65$